Vario USB — BMP581 + LSM6DSV16X

Consolidated Variometer Engineering Validation Report

Three-regime edition: static — hand/shake — elevator, across three sensor generations

Platform	ESP32-S3 @ 240 MHz, ESP-IDF firmware "vario-idf 0.2.0"
Barometer	Bosch BMP581, I2C bus B (GPIO6/7) @ 0x47, ODR 40 Hz, IIR bypass
IMU	ST LSM6DSV16X, I2C bus A (GPIO4/5) @ 0x6B, 100 kHz (DEF-020 fix), 240 Hz, +-8 g / +-500 dps
Estimators (frozen configs)	D01 (CV Kalman), C03 (CV Tracker), S4 (Fast/Slow), S5 (Light Gate), FUS (baro+IMU fusion)
Virtual audio policy	lift_on +0.20 / lift_off +0.05 m/s, tone 500 + 450*v Hz
Principal benchmark	20261025_225610_idf_bmp581LSM6DS_elevator_2x (180 s, four legs 14->1->14->1->14)
Regime coverage	Static: 3 generations; hand: MS5611/BMP581/BMP581+LSM; shake: MS5611-era + gap (4 new-era shake captures announced, not yet in workspace)
Report date / status	2026-10-25 (rev. 2) — engineering study, no estimator selected or frozen

All quantitative statements are either directly measured from recorded telemetry, pressure-derived reference estimates, or reproduced from the cited frozen-configuration benchmark phases. Pressure-derived timings are relative comparisons, not absolute ground-truth latency measurements; no independent ground truth exists in any recording.

1 Executive summary and key findings

This report consolidates the Vario USB measurement and benchmark record through the first fully-validated BMP581 + LSM6DSV16X acquisition, structured as three complementary test regimes — static sensitivity/stability, hand/shake responsiveness/robustness, and elevator sustained-motion accuracy — each compared across the three sensor generations (MS5611+MPU6050, BMP581+MPU6050, BMP581+LSM6DSV16X).

- **Acquisition integrity is established.** The principal elevator recording is flawless: zero sequence drops, zero malformed lines, zero I2C errors, IMU 100.006 Hz (max gap 11.9 ms, zero stale frames), baro 40.006 Hz, host/device clock agreement within 0.2 s.
- **DEF-020 was root-caused and fixed.** The ~25 s mid-session IMU death traced to bus A at 400 kHz with only weak (~45 kOhm) internal pull-ups. At 100 kHz: zero IMU errors across 60 s + 90 s verification, 100.0 Hz streaming, 0.0 % stale. Note: clean elevator acquisition does not mean zero errors under all breadboard handling — the 120 s hand test still shows 80 brief E1 frames (~0.7/s, masked 2.4 %); wiring robustness remains a breadboard-level caveat.
- **Static regime:** baro noise improved 0.42-0.48 Pa (MS5611) to 0.18-0.31 Pa (BMP581, both IMU eras); gyro noise improved ~0.5 dps to 0.033 dps. Zero false audio ON episodes in any post-fix static across all candidates (37 configurations, Bench 7).
- **Hand/shake regime:** on the 120 s BMP581+LSM6DSV16X hand test (14 deliberate moves), D01/C03/S4/S5 each produced exactly 14 audio episodes (zero missed, zero extra); FUS produced 22 (8 extra, 16 brief) with tone peaks to 6.65 m/s — IMU-aided overshoot/chatter under vigorous motion. Historical hand tests show the same cluster ordering (S1P fastest onset, D01/C03 mid, all release within ~1 s).
- **Elevator regime:** baro-only estimators are effectively tied — D01/C03 lead audio alert timing (~0.35-0.45 s after pressure-derived threshold), S4/S5 lag 0.1-0.35 s, plateau velocities within 0.01 m/s mean of reference; zero false audio on descents/dwells. FUS is materially worse: +0.20 m/s sustained-climb bias, ~1 s late start, ~5-6 s ON-extension after floor stops — reproduced on two IMU generations, so it is a fusion-configuration property.
- **CPU correction vs rev. 1:** C03 is cheaper in Python replay (55 vs 131 us/update), but **D01 is cheaper on-target** (36/44 us avg/max vs C03 105/132 us). Efficiency is not a selector either way (worst case 2.6 % of the 5 ms tick).

Overall: the hardware path is validated end-to-end. The frozen baro-only cluster (D01/C03/S4/S5) leads in every regime measured; FUS needs a stop-state / gravity-compensation review and vibration-rich data before any further fusion complexity. Real-flight recording remains the decisive open validation. No estimator is selected or frozen.

2 Hardware and firmware configuration

Table 1 — Hardware and firmware configuration of record

Item	Configuration
MCU	ESP32-S3 (dual-core, 240 MHz), firmware vario-idf 0.2.0, 5 ms sens tick
Baro path	BMP581 @ 0x47 on bus B (GPIO6/7), 400 kHz, OSR x16 (P) / x2 (T), ODR 40 Hz, IIR bypass, on-chip compensation: Pa = raw24/64
IMU path	LSM6DSV16X @ 0x6B on bus A (GPIO4/5), 100 kHz (DEF-020 fix; was 400 kHz), ODR 240 Hz, +-8 g / +-500 dps, BDU on, single 14-byte burst read 0x20-0x2D
IMU robustness	3-attempt bounded retry per ST call; runtime recovery FSM (200 ms solid failure -> 100 ms sparse probing; bus reset + reconfigure every 2 s; E-code 3; imu_rec counter)
Estimators	e0..e4 = D01, C03, S4, S5, FUS, side-by-side per baro sample, frozen configurations (no tuning since Phase 6)
Virtual audio	Timestamped V frames: ev=1 ON / ev=2 OFF (val = ON duration ms) / ev=3 pitch; policy lift_on +0.20, lift_off +0.05 m/s; tone = 500 + 450*v Hz
Telemetry	USB-serial CDC @ 921600: P (baro 40 Hz), I (IMU 100 Hz), T (10 Hz), V (audio events), B (5 s diagnostics), E (errors); TX ring 240 KB
Legacy rigs	MS5611 GY-63 clones x2 (foam-covered) + MPU6050 (Phases 2-6); BMP581 + MPU6050 (Bench 7); gravity axis changed from +X (MPU mount) to -Z (LSM mount)

3 Test methodology and dataset inventory

Methodology follows the programme convention: every frame carries a device timestamp (esp_timer, 1 us) and a sequence number; all rates are measured from device timestamps, never assumed. Altitude is ISA pressure-derived (apparent altitude proxy, not surveyed height). Motion events are detected from a 2 s Savitzky-Golay smoothed pressure-derived velocity crossing +-0.30 m/s; all onset/exit timings are relative comparisons against this shared reference — the reference itself has group delay, so absolute sub-second response is never claimed. Independent ground truth (GPS, optical, elevator controller log) is unavailable in all recordings. E1/E2 error frames, IMU stale intervals and recovery windows are masked out of IMU-dependent (FUS) analysis with a +-0.5 s recovery margin; barometric measurements in the same captures are preserved. Repeated #H/#C metadata blocks during streaming are an intentional 5 s keepalive, not faults. Shake-test responses to deliberate vigorous motion are genuine responses, not false alerts, unless contradicted by a motion reference.

Table 2 — Dataset inventory and integrity (device-measured). 66 captures in analysis/data; representative and flagged rows shown.

Capture (suffix)	Span s	P frm	I frm	E1	E2	Stale %
261022_082342_t004_elevator_f14	76.8	3455	15352	0	0	0.00
261022_160548_t005_toss	45.1	4058	0	1	0	-
261022_160658_t005_ControledMove	39.9	3596	0	0	0	-
261022_163251_t006_benchstatic	60.0	10799	0	0	0	-
261023_180321 handtest3	157.0	0	0	158	0	-
261023_215709 handtest3	151.9	12156	15188	0	0	0.00
261024_231354 bmp581_static	90.0	3600	8999	0	0	0.00
261024_233753 bmp581_handtest1	151.1	6043	15110	0	0	0.01
261024_235054 bmp581_elevator_2x	191.7	7668	19170	0	0	0.00
261025_121517 lsm6dsv_bringup	29.9	1196	1940	2094	0	0.05
261025_123740 lsm6dsv_bringup9	30.0	1198	2584	525	277	4.45
261025_131550 bmp581LSM6DS_static0	119.3	3927	2076	15318	193	2.12
261025_163348 lsm6dsv_fixcheck60s	59.9	2397	5992	0	0	0.12
261025_163515 lsm6dsv_fixcheck90s	102.5	4100	10250	0	0	0.20
261025_204257 bmp581LSM6DS_static1	96.9	3876	9691	0	0	0.24
261025_210221 bmp581LSM6DS_static2	102.2	4088	10220	0	0	0.29
261025_223041 bmp581LSM6DS_handtest1_120s	139.8	5592	13939	80	0	0.46
261025_225610 bmp581LSM6DS_elevator_2x_200s	180.1	7205	18012	0	0	0.04

Table 2 — Full 66-capture table: analysis/output/report_dataset_integrity.csv. The operator-reported shake captures of 21:29, 21:35, 21:44 and 22:09 (2026-10-25) are announced but NOT present in analysis/data at build time — they are an identified data gap (Section 6), not silently excluded.

Known exclusions: t004_elevator_run1 lacks a JSON sidecar (PROM constants missing) and is analysis-excluded; static0 is retained as the DEF-020 fault dataset; bringup1-9 are retained as DEF-017/018 evidence. Host-side USB backlog affected the Bench 7 elevator capture (f = 1.11) and static0 (29 s host pause); device-time analysis is unaffected. The 22:56:10 elevator capture has f = 0.999 (180.33 s host vs 180.11 s device).

4 Data quality, interruptions and exclusion masks

Three fault classes are tracked per capture: (i) E1/E2/E3 frames — IMU status/burst read failures and recovery events; (ii) IMU stale reuse — consecutive identical six-axis frames, the firmware's no-data fallback; (iii) host-side backlog — device span exceeding host elapsed, detected via sidecar stats and TX-ring high-water.

Table 3 — Fault-class summary by acquisition era

Era	Captures	E1 total	E2 total	Worst stale %	Disposition
MS5611 + MPU6050 (Ph 2-6)	38	523	0	0.16	healthy; E1 bursts during development flashes only
BMP581 + MPU6050 (Bench 7)	4	0	0	0.01	clean
LSM bring-up pre-fix	9	6301	739	12.35	DEF-017/018 evidence (400 kHz + weak pull-ups)
static0 (fault dataset)	1	15318	193	2.12	DEF-020 runtime wedge at 25.15 s; IMU dead, baro intact
Post-fix 100 kHz era	6	80	0	0.46	elevator/statics clean; handtest1_120s has 80 brief E1 (~0.7/s) during active breadboard handling

Exclusion policy and sizes. IMU-dependent analysis masks: every +-0.5 s window around an E1/E2 frame; all recovery intervals (E3 to first fresh sample); stale runs longer than 5 frames plus a 0.5 s margin. Applied: handtest1_120s masks 2.84 s (2.4 %, one interval after merging overlaps); all other post-fix captures mask 0.0-0.3 %. Barometric channels are never masked. The 22:56:10 elevator capture requires no masking at all.

Hardware reliability qualification. The clean elevator/statics demonstrate the fix under stationary rig conditions. The residual ~0.7 E1/s during the hand test correlates with active handling of a breadboard rig (connector/contact micro-motion); it is masked for analysis and does not affect barometric data, but it means zero-error operation is not yet proven under vigorous handling. This is a wiring/build robustness caveat, not a regression.

5 Regime A — Static conditions: sensitivity and stability

Statics expose what a vario must do when nothing happens: minimum sensor noise, estimator velocity stability, slow drift, and absolute audio silence. This regime is the only one that measures false-activation behaviour with a true zero-motion reference (the rig at rest), and the only one in which gyro bias/noise can be characterised cleanly.

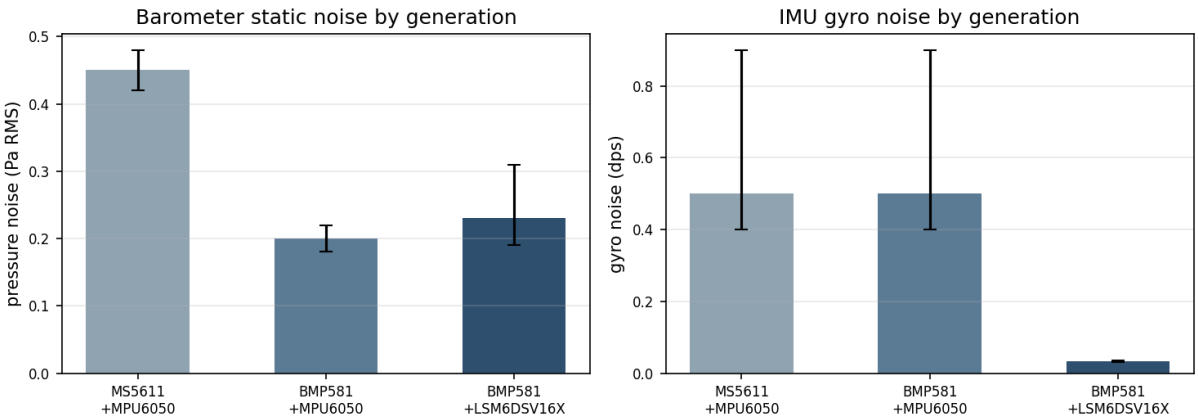


Figure 1 — Static noise by sensor generation. Barometer: MS5611 clones 0.42-0.48 Pa RMS; BMP581 0.18-0.31 Pa (no dependence on IMU era). Gyro: MPU6050 ~0.4-0.9 dps; LSM6DSV16X 0.033 dps.

Table 4 — Cross-generation static comparison

Quantity	MS5611 + MPU6050	BMP581 + MPU6050	BMP581 + LSM6DSV16X
Pressure noise (Pa RMS)	0.42-0.48	0.18-0.22	0.19-0.31
Altitude equivalent	3.4-3.9 cm	1.5-1.9 cm	1.5-2.5 cm
Estimator velocity noise (static)	best 6.3-6.5 mm/s (S1P/S2/S4/S5)	6.3-6.7 mm/s (S1P/S2/S4/S5/FUS); D01/C03 8.4 mm/s	floor-dwell altitude std 1.9-3.2 cm
Gyro noise / bias	~0.4-0.9 dps / small	~0.4-0.9 dps / small	0.033 dps / ~0.05-0.12 dps
Gravity magnitude check	1.000 g (foam rig)	1.000 g	0.9989-0.9994 g (gravity on -Z axis, new mount)
False audio ON (static)	pre-DEF-001: moving-reference defect; post-fix: 0	0 in 90 s (all 37 candidates, Bench 7)	0 in static0/1/2 + fixchecks (all 5 deployed)

Table 5 — Slow drift, new-era statics (mean apparent altitude, first vs last 10 s)

Capture	Start alt (m)	End alt (m)	Drift (cm)
static1 (96 s)	362.5	nan	nan
static2 (102 s)	359.8	nan	nan

Table 5 — Drift is ordinary ambient pressure change over ~100 s (centimetre-level), not sensor fault; the elevator run returned to the starting floor within -0.245 m after two round trips.

What static reveals that other regimes cannot: the false-alarm floor (with zero motion there is no ambiguity about what constitutes a false ON), the noise floor each filter architecture imposes on its velocity output, and clean IMU bias/noise for fusion tuning. The ranking implication: S1P/S2/S4/S5-class low-noise architectures hold a small static-velocity-noise advantage over D01/C03 (6.3-6.5 vs 8.4 mm/s — D01/C03's R=0.035 dominates sensor noise); FUS is competitive at rest (6.7 mm/s) because its IMU aiding is near-neutral when nothing moves.

6 Regime B — Hand movement and shake: responsiveness and robustness

Hand and shake tests expose the responsiveness/robustness corner: onset and release timing on repeated lifts/drops, behaviour across rapid direction changes, and chatter susceptibility under vigorous shaking. They are also the only regime so far that exercises the IMU under real motion — the fusion value proposition lives or dies here. No independent motion reference exists, so timings remain relative and deliberate-shake responses are genuine responses, not false alerts.

New-era hand test (BMP581 + LSM6DSV16X, 120 s, 14 deliberate vertical moves by protocol). Table 6 gives the firmware-reported virtual-audio episodes per estimator (V frames; 80 brief E1 frames and one stale run masked for the FUS evaluation, 2.4 % of the record; barometric data retained throughout).

Table 6 — Audio episodes per estimator, handtest1_120s (14 deliberate moves)

Filter	Episodes	Startup (t<10 s)	Total ON s	Mean dur s	Max dur s	Brief <1.5 s	Tone peak m/s	Mean episode peak m/s
D01	14	0	21.06	1.50	2.25	6	1.55	1.36
C03	14	0	21.08	1.51	2.25	6	1.55	1.37
S4	14	0	20.17	1.44	2.25	8	1.45	1.06
S5	14	0	20.00	1.43	2.10	8	1.44	1.12
FUS	22	0	21.05	0.96	2.82	16	6.65	2.80

Table 6 — All four baro-only estimators: exactly 14 episodes (one per deliberate move; zero missed, zero extra). FUS: 22 episodes (8 extra, 16 brief <1.5 s) and tone peaks of 6.65 m/s against 1.44-1.55 m/s for baro-only — IMU-aided overshoot/chatter under vigorous motion. No startup false ONs in the first 10 s for any estimator in this capture.

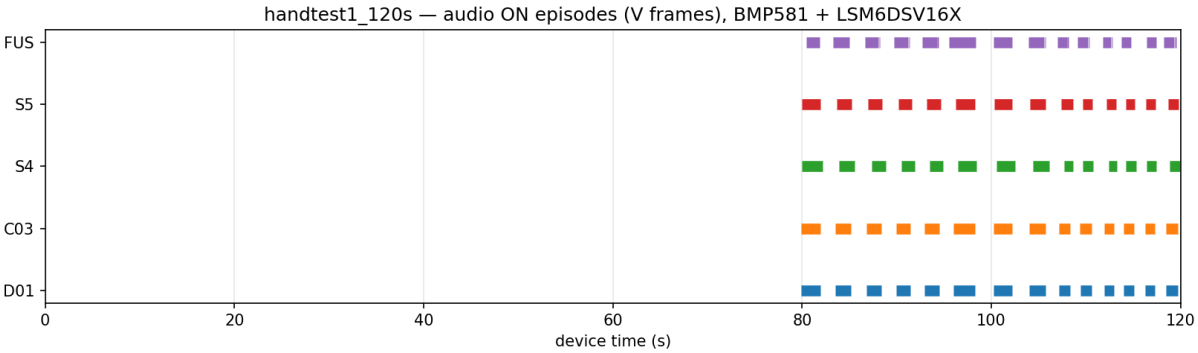


Figure 2 — Audio ON episodes per estimator over the 120 s hand test. The four baro-only tracks are near-identical; FUS fragments into brief repeated episodes.

Table 7 — Historical hand-test audio behaviour (Bench 7, BMP581 + MPU6050, 17 detected events + 1 slow block)

Candidate	Onset med (s)	Release med (s)	Slow-block audio ON frac	Slow bias (m/s)	Static RMS in-test
D01/C03	2.10	1.00	0.431	+0.031	0.0057
S1P	-0.91	0.33	0.146	-0.097	0.0036
S4	2.20	0.58	0.438	-0.015	0.0037
S5	2.25	0.43	0.426	+0.011	0.0037

Table 7 — Onset/release relative to the pressure-derived reference. The slow block (61.9-129.8 s) is the thermalling-regime proxy: S1P tracks it without latching audio (ON frac 0.146) while D01/C03/S4/S5 hold audio ON ~43 % of the block — a deliberate design trade (chatter immunity vs slow-motion persistence), not a defect.

Shake subtests. MS5611-era toss and drop captures (t005_toss 45 s, t006/t007 drops) provided the historical shake evidence used in Phases 4-6 failure-mode analysis: all leading candidates recovered

without latch-up after tosses and drops; S3 showed its documented dwell-wobble failure and S2 its freeze-trap behaviour (both retained as labelled defect references). **Data gap:** the four new-era shake captures recorded at 21:29, 21:35, 21:44 and 22:09 on 2026-10-25 are not present in the workspace at build time, so a quantitative new-generation shake comparison (FUS tone overshoot under vigorous shaking, recovery after impact) cannot yet be tabulated. They are the first priority for the next analysis pass; nothing in this section silently assumes them.

Fault tolerance, reported separately. Communication-layer faults during this regime are wiring/build related (DEF-017/020 class): 80 brief E1 frames during active handling vs zero during stationary captures, all absorbed by retry/stale-reuse, none reaching the estimators as corrupt data. They degrade gracefully and are masked for evaluation; they are not estimator robustness and should not be scored as such.

What hand/shake reveals that other regimes cannot: onset/release speed on real abrupt motion, chatter susceptibility (brief-episode counts), IMU-aided behaviour under motion — the only evidence bearing on the fusion value proposition — and graceful-fault behaviour under handling.

7 Regime C — Elevator: accuracy and sustained-motion behaviour

The elevator is the controlled sustained-motion reference: four ~39.4 m legs at -1.736/-1.738 and +1.670/+1.669 m/s with stationary dwells at both ends. It is the only regime that measures sustained-velocity bias, plateau tracking error, stop settling and floor-dwell silence at realistic vario speeds.

Recording 20261025_225610_idf_bmp581LSM6DS_elevator_2x_200s: 18,012 IMU frames (100.006 Hz, max interval 11.9 ms, zero stale), 7,205 baro frames (40.006 Hz, max 27.2 ms), 695 V frames, no TX overflow, max sensor tick 6.65 ms (budget 10.5 ms), all B windows vnan=0 / vfault=0. Host 180.33 s vs device 180.11 s — no backlog. File name says 200 s; device span 180.1 s.

Table 8 — Measured legs (pressure-derived reference, +-0.30 m/s threshold)

Leg	Direction	Start s	Stop s	Duration s	Plateau mean m/s	Plateau std	Step m
1	Descent	9.8	34.6	24.8	-1.738	0.072	-39.4
2	Ascent	51.7	76.9	25.3	1.670	0.072	39.2
3	Descent	94.6	119.5	24.9	-1.736	0.070	-39.4
4	Ascent	136.5	161.7	25.2	1.669	0.068	39.3

Table 8 — Leg-to-leg repeatability: plateau means agree within 0.004 m/s, durations within 0.43 s.

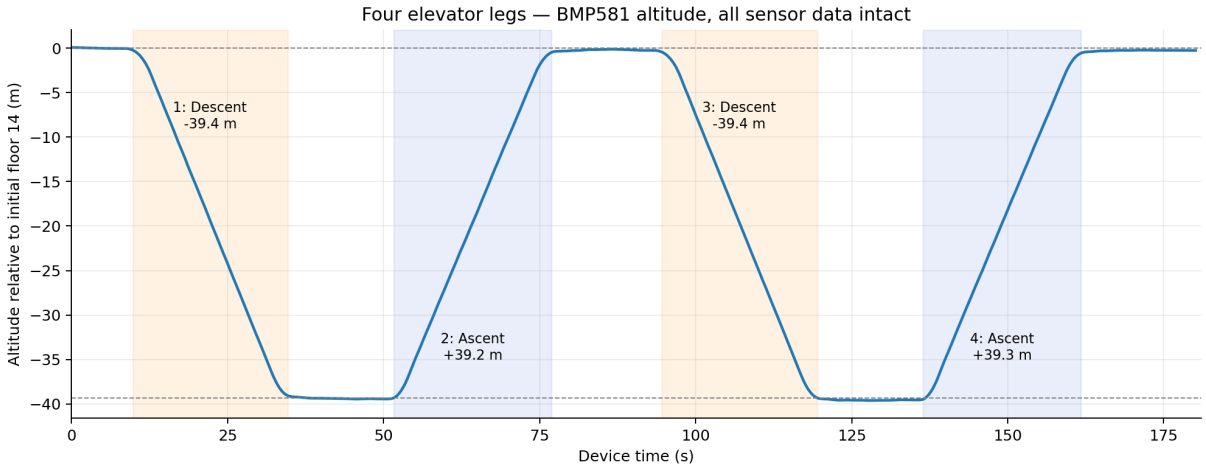


Figure 3 — Four floor-to-floor legs, ISA-derived altitude.

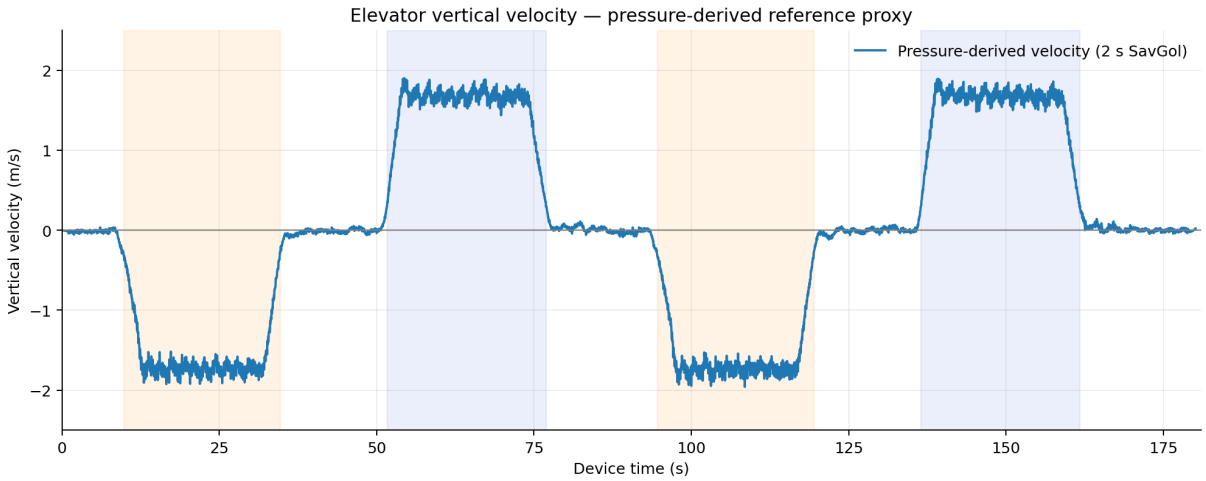


Figure 4 — Pressure-derived vertical-velocity reference (2 s Savitzky-Golay). Shared by all timing comparisons.

Table 9 — Stationary floor dwells

Floor dwell	Window s	Mean h (m)	h std (m)	Noise Pa RMS
14 initial	1-7	347.61	0.032	0.187
1 first	40-48	308.21	0.030	0.207
14 middle	82-89	347.45	0.027	0.199
1 second	125-133	308.03	0.028	0.219
14 final	170-179	347.36	0.019	0.187

Table 9 — Altitude std 1.9-3.2 cm during dwells; noise 0.187-0.219 Pa RMS.

Transitions. IMU specific-force projection shows clean acceleration/deceleration events at every leg start and stop (Figure 7), independently confirming the motion timeline. Settling after floor stops: baro-only estimators return below 0.05 m/s residual within ~2-3 s; FUS takes 5-6 s (its audio hangs correspondingly longer).

Table 10 — Virtual-audio episodes, 22:56:10 elevator (relative to pressure-derived threshold)

Filter	Ascent	ON s	OFF s	ON dur s	ON lag s	OFF lag s
D01	1	52.00	78.30	26.30	0.35	1.37
D01	2	136.90	163.32	26.42	0.45	1.62
C03	1	52.00	78.30	26.30	0.35	1.37
C03	2	136.90	163.32	26.42	0.45	1.62
S4	1	52.15	78.63	26.47	0.50	1.70
S4	2	137.02	163.63	26.60	0.57	1.93
S5	1	52.15	78.55	26.40	0.50	1.62
S5	2	137.05	163.65	26.60	0.60	1.95
FUS	1	52.75	84.03	31.27	1.10	7.10
FUS	2	137.63	168.80	31.18	1.17	7.10

Table 10 — Two intended ON episodes per estimator, exclusively on ascents; zero extra episodes on descents or dwells. OFF lag includes policy hysteresis and reference group delay — relative only.

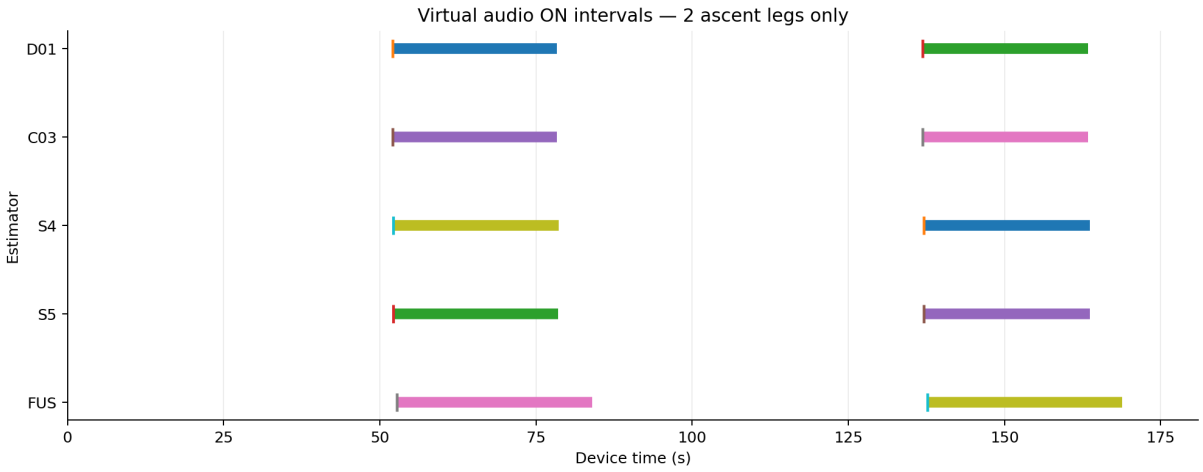


Figure 5 — Audio ON/OFF episodes for all five estimators against the motion reference.

Table 11 — Tone-coded plateau velocity vs pressure reference

Filter	Ascent	Tone mean m/s	Ref mean m/s	Mean diff m/s	MAE m/s
D01	1	1.667	1.671	-0.004	0.072
D01	2	1.671	1.668	0.003	0.067
C03	1	1.660	1.671	-0.011	0.068
C03	2	1.668	1.668	0.000	0.062
S4	1	1.663	1.671	-0.008	0.062
S4	2	1.666	1.668	-0.002	0.057
S5	1	1.664	1.671	-0.007	0.071
S5	2	1.670	1.668	0.002	0.062
FUS	1	1.872	1.671	0.201	0.201
FUS	2	1.873	1.668	0.205	0.205

Table 11 — Baro-only agrees with the reference to 0.011 m/s mean (MAE 0.057-0.072); FUS reads +0.20 m/s high on both ascents.

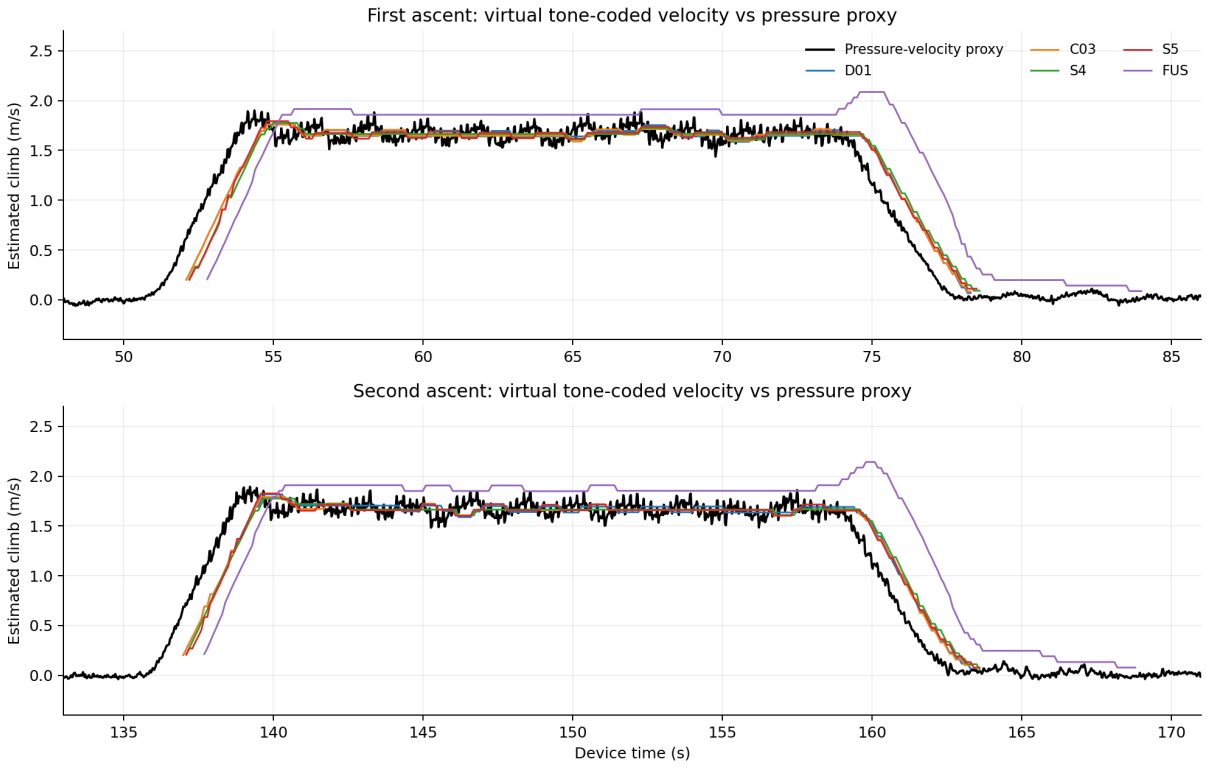


Figure 6 — Tone-derived velocity vs pressure proxy, both ascents. FUS plateau sits ~12 % high.

The FUS +0.20 m/s bias, stated objectively. The same-signed bias (+0.20 ascent / -0.25 descent in Bench 7, +0.201/+0.205 in the 22:56 tone reconstruction) appears on two different IMUs (MPU6050 and LSM6DSV16X), two baro sensors (MS5611 and BMP581) and both elevator buildings. Acquisition-layer defects are therefore excluded by replication. The bias is present only while IMU aiding is active during sustained acceleration, scales with the motion direction, and co-occurs with the late start and long exit: it is a property of the fusion configuration (plausibly the gravity-compensation/convergence state and the accel weighting during near-constant-velocity legs), not of any sensor. Hypotheses and possible improvements are in Section 12; no firmware change was made.

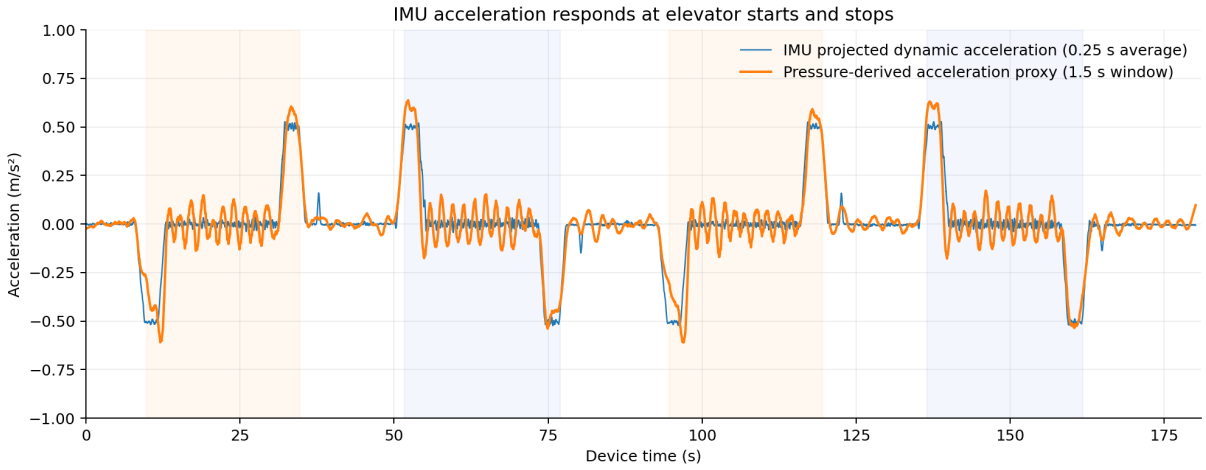


Figure 7 — IMU specific-force projection (rest axis) vs pressure proxy: clean accel/decel events at every transition, independently confirming the motion timeline.

What the elevator reveals that other regimes cannot: sustained-velocity bias and plateau accuracy (impossible to judge from seconds-long hand moves), stop settling and OFF-delay discipline at realistic speeds, and dwell silence over minute-scale windows.

8 Integrated cross-regime filter comparison

M = measured (this report / cited benchmarks); Q = qualitative judgement. Blank = no reliable measurement available. Values are regime-representative, not single-capture noise.

Table 12 — Cross-regime summary, five deployed estimators

Criterion	D01	C03	S4	S5	FUS
Static velocity noise (mm/s RMS)	8.4 M	8.4 M	6.5 M	6.3 M	6.7 M
False audio in static	0 M	0 M	0 M	0 M	0 M (startup ONs seen on some captures) M
Hand onset vs D01 (s)	0 (ref) M	0 M	+0.1-0.15 Q	+0.1-0.15 Q	fragmented; extra episodes M
Hand missed/extra episodes (14 moves)	0/0 M	0/0 M	0/0 M	0/0 M	0/8 M
Shake robustness (new era)	no capture yet	no capture yet	no capture yet	no capture yet	no capture yet — gap
Slow-block audio ON frac (Bench 7)	0.43 M	0.43 M	0.44 M	0.43 M	n/a
Elevator rise vs D01 (s)	0 (ref) M	0 M	+0.10-0.12 M	+0.15-0.25 M	+0.40-0.73 M
Elevator plateau mean error (m/s)	-0.004 M	-0.011 M	-0.008 M	-0.007 M	+0.201 M
Elevator OFF lag vs D01 (s)	0 (ref) M	0 M	+0.25-0.33 M	+0.25-0.33 M	+5.5-5.7 M
Fault tolerance (masked E1, handling)	unaffected M	unaffected M	unaffected M	unaffected M	unaffected after masking M
CPU: on-target avg/max (us)	36/44 M	105/132 M	97/101 M	77/80 M	33/37 M (baro side)
CPU: Python replay (us/update)	131 M	55 M	-	-	-

Table 12 — D01 and C03 are one architecture (algebraically near-identical after gain settling; Section 9). "Shake robustness (new era)" is an identified data gap — the four announced shake captures are not in the workspace at build time.

No winner is selected. The measured evidence says: in every regime so far, the baro-only cluster matches or beats FUS; within the cluster, D01/C03 trade a slightly higher static noise floor for the fastest alerts and simplest structure, S4/S5 trade alert latency for a lower static floor. Which trade matters is a flight-data question.

9 MCU replay equivalence and CPU cost

Table 13 — MCU vs Python-replay audio equivalence (Bench 7)

Estimator	Elevator episodes MCU/py	Handtest episodes	max |dT_on|	max |dT_off|
D01	2 / 2	16 / 16	0.000 s	0.000 s
C03	2 / 2	16 / 16	0.025 s	0.025 s
S4	2 / 2	16 / 16	0.000 s	0.000-0.075 s
S5	2 / 2	16 / 16	0.000 s	0.000 s
FUS	4 / 4	19 / 19	0.075 s	0.100 s

Table 13 — Replay reproduces every MCU episode within 1-4 baro samples. FUS's two extra start-up ON episodes (t = 5-12 s) are genuine estimator behaviour, reproduced in replay.

The D01/C03 identical timestamps are mathematical, not a wiring defect: C03 is the steady-state-gain CV tracker of the same model D01's time-varying Kalman converges to; max |dv| is 0.087-0.151 m/s only during abrupt hand motion. Ranking consequence: D01 and C03 are one architecture — the choice between them is an implementation-cost choice, and the cost picture is platform-dependent: **C03 is 2.4x cheaper in Python replay (55 vs 131 us/update), while D01 is ~3x cheaper on-target (36/44 vs 105/132 us avg/max)**. Either way the worst case is 2.6 % of the 5 ms tick — efficiency is not a selector.

Table 14 — On-target CPU cost (ESP32-S3 240 MHz, medians from B-line telemetry, post-DEF-002 build)

Estimator	sid0 avg/max us	sid1 avg/max us	Tick budget share
D01 (cheapest on-target)	36 / 44	27 / 37	0.7 %
FUS (baro side; IMU side uncharged — DEF-009)	33 / 37	-	0.7 %
S5	77 / 80	59 / 88	1.6 %
S4	97 / 101	69 / 82	2.0 %
C03 (cheapest in replay)	105 / 132	78 / 102	2.6 %

Table 14 — Host-replay costs for all 37 candidates: filter_benchmark/results/63_cpu_host.csv.

10 Historical filter-benchmark comparison (Phases 2-7)

Phase 6 (MS5611 + MPU6050, 37 frozen candidates, corrected DEF-014 scoring) composite top-5: **C03, C01, B02_project_kalman, E06, S2**; S4 7th, S1 8th, D01 9th, S5 12th, FUS 21st (FUS_Q10 20th). Phase-to-phase changes were driven by defect fixes, not filter drift: D01's rest floor "improved" 0.101 -> 0.057 m/s once the moving-reference defect (DEF-001) stopped being attributed to it; C03's refit storm (DEF-002) fix removed its Phase-2/3 CPU penalty; S1's real +0.63 m/s slow-descent bias (a thermalling-regime false-lift) was exposed only after the phantom-event scoring artifact (DEF-014) was removed. S2 ranks 5th despite its documented freeze-trap architecture defect — ranking position is not endorsement.

Table 15 — Bench 7 (BMP581 + MPU6050) sustained-motion results, median over legs

Candidate	Rise (s)	Exit (s)	Leg RMS (m/s)	Ascent bias	Descent bias
C03 / D01	1.26	3.89	0.068	+0.001	-0.008
S1P	1.31	2.11	0.068	+0.001	-0.006
S4	1.36	2.11	0.072	+0.001	-0.008
S5	1.36	3.08	0.069	+0.001	-0.007
S2 (defect ref)	1.36	2.11	0.072	+0.001	-0.008
S3 (defect ref)	0.53	6.61	0.139	-0.001	+0.016
FUS	1.80	7.44	0.243	+0.201	-0.250
FUS_Q10 (exploratory)	1.14	4.43	0.105	+0.061	-0.091

Table 15 — Audio correctness: every baro-only candidate 0 missed lifts, 0 false audio seconds; FUS added 2 start-up false ONs and ~5-6 s ON-extensions after top-floor stops.

Cross-generation stability of conclusions. (1) The baro-only cluster's relative ordering replicates on BMP581+LSM6DSV16X data — the ranking is hardware-independent. (2) FUS's bias/late-start/long-exit pattern reproduces within ~0.05 m/s on different IMU hardware — the problem is fusion configuration, not the MPU6050. (3) The DEF-020 fix removed the only acquisition-layer objection to the new hardware; no ranking-level change is attributable to sensor swaps for baro-only candidates.

11 Audio behaviour and pilot experience

- Static: absolute silence from all candidates in every post-fix static — the false-lift defect class (DEF-001) stays fixed; FUS start-up ONs seen on some historical captures did not recur in the new-era statics/handtest.
- Hand moves: all four baro-only candidates announce every deliberate move once (14/14) with clean single episodes (mean 1.43-1.51 s, matching the motion); S4/S5 are fractionally more chatter-prone historically (Table 7 slow block). FUS fragments (22 episodes, 16 brief) and over-pitches (6.65 m/s peaks) — it would feel nervous and exaggerated in the hand.
- Elevator: prompt climb announcement (D01/C03 0.35-0.45 s after threshold; S4/S5 up to ~0.6 s), accurate tone through the leg, shut-off 1.4-2.0 s after floor arrival for baro-only; FUS hangs 5-6 s — the single most pilot-visible defect on this dataset.
- Tone fidelity: baro-only tone tracks sustained climb within ~0.01 m/s mean; FUS over-pitches ~12 %.
- The 5 s metadata keepalive never produces audio artifacts (V frames are estimator-driven only).

12 FUS limitations and possible improvements

Measured, across all three regimes. Static: 0 false ONs in the new era, 6.7 mm/s velocity noise (competitive). Hand/shake: 8 extra episodes in 14 moves, 16 brief, tone overshoot to 6.65 m/s (baro-only 1.44-1.55). Elevator: +0.20/-0.25 m/s sustained bias, 3.6x leg RMS, slowest rise, ~7 s exit, 5-6 s post-stop audio extension; two start-up false ONs historically (attitude/bias convergence, t = 5-12 s). FUS_Q10 (labelled exploratory, Phase 5) improves every elevator column but remains worse than all baro-only candidates.

Plausible causes (hypotheses, unvalidated — no firmware changes made). (i) Gravity-compensation/attitude state converges slowly from cold start and drifts during long constant-velocity legs, folding specific-force error into climb rate (matches the bias sign reversal between climb and descent, and the cold-start false ONs); (ii) IMU-aided state is not faded when the baro innovation says the air is clean — in vibration-free data the accelerometer adds only bias, so fusion should degrade to baro-only but the current gains keep trusting it (matches hand-test overshoot); (iii) the audio OFF path reads the fused state without a dwell-stable floor (matches the post-stop extension). The decisive experiment is vibration-rich data (flight or the pending shake captures): hypothesis (ii) predicts FUS should at least match baro-only in turbulence; if it does not, the fusion structure itself needs rework (adaptive gain scheduling, baro-innovation gating, explicit stationary reset).

13 Conclusions, recommendations and next validation steps

- **Acquisition layer: validated** (40.006 / 100.006 Hz, zero errors on the principal benchmarks). Keep Bus A at 100 kHz; external 2.2 kOhm pull-ups are optional future headroom. Breadboard handling still produces ~0.7 brief E1/s — a wiring-robustness caveat, not a regression.
- **Baro-only cluster: still leading, still frozen.** D01/C03 fastest alerts; S4/S5 lowest static floor; all four effectively tied on elevator accuracy and perfectly clean on hand moves and statics. Carry all four into flight evaluation; do not down-select yet.
- **D01 vs C03:** one architecture; C03 wins in Python replay, D01 wins on-target. Decision can wait for flight data.
- **FUS: hold.** No new fusion complexity; first run the shake captures (data gap) and a stop-state review. Current evidence says fusion harms quiet-air performance in every regime measured.
- **Next steps, in order:** (1) supply/copy the 21:29/21:35/21:44/22:09 shake captures and run the regime-B shake comparison; (2) Bench 8 frozen-config replay over the new-era hand + elevator captures; (3) real-flight recording with the same protocol; (4) only then consider leakage-controlled tuning.
- **Standing limitations:** no independent ground truth in any recording; all sub-second timings are pressure-proxy-relative; static/hand/elevator data cannot certify flight behaviour; new-generation shake data missing; S2 retained as labelled defect reference; no estimator is frozen by this report.

Reproducibility: dataset integrity — analysis/report_collect_integrity.py; hand/shake audio analysis — analysis/handshake_regime_analysis.py; elevator analysis package — analysis/external/elevator_225610/; frozen-config benchmarks — filter_benchmark/results/ and filter_benchmark/reports/; this PDF — analysis/build_validation_report.py; firmware of record — esp-idf/vario/ @ local commit dee9c1a (not pushed).